

PPA

Title: Pressure-Sensing Mobility Aid with Data Interface

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Pressure-Sensing Mobility Aid with Data Interface

ABSTRACT

A sensorized orthopedic device tip is disclosed comprising a polyurethane rubber base, an integrated load cell, and a Bluetooth microchip configured to replace conventional tips on crutches, canes, walkers, and similar aids. The load cell measures user-applied load to assess adherence to prescribed partial or full weight-bearing, enabling optimization of recovery and treatment recommendations. Measured data are wirelessly transmitted via Bluetooth to a software application for review by clinicians, facilitating remote monitoring, data-driven adjustment of rehabilitation protocols, and delivery of individualized treatment recommendations to patients.

FIG. 1

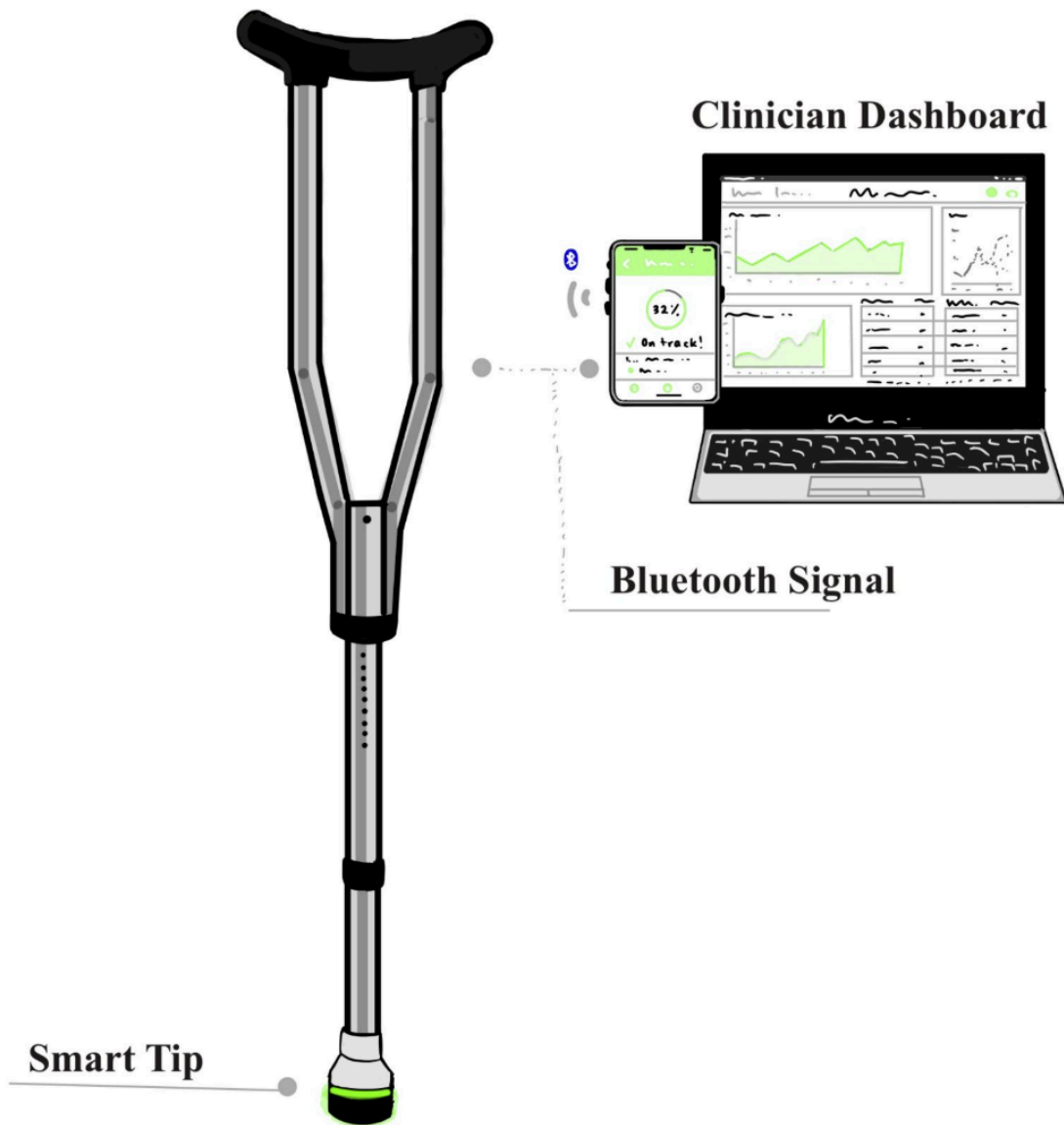


FIG. 2

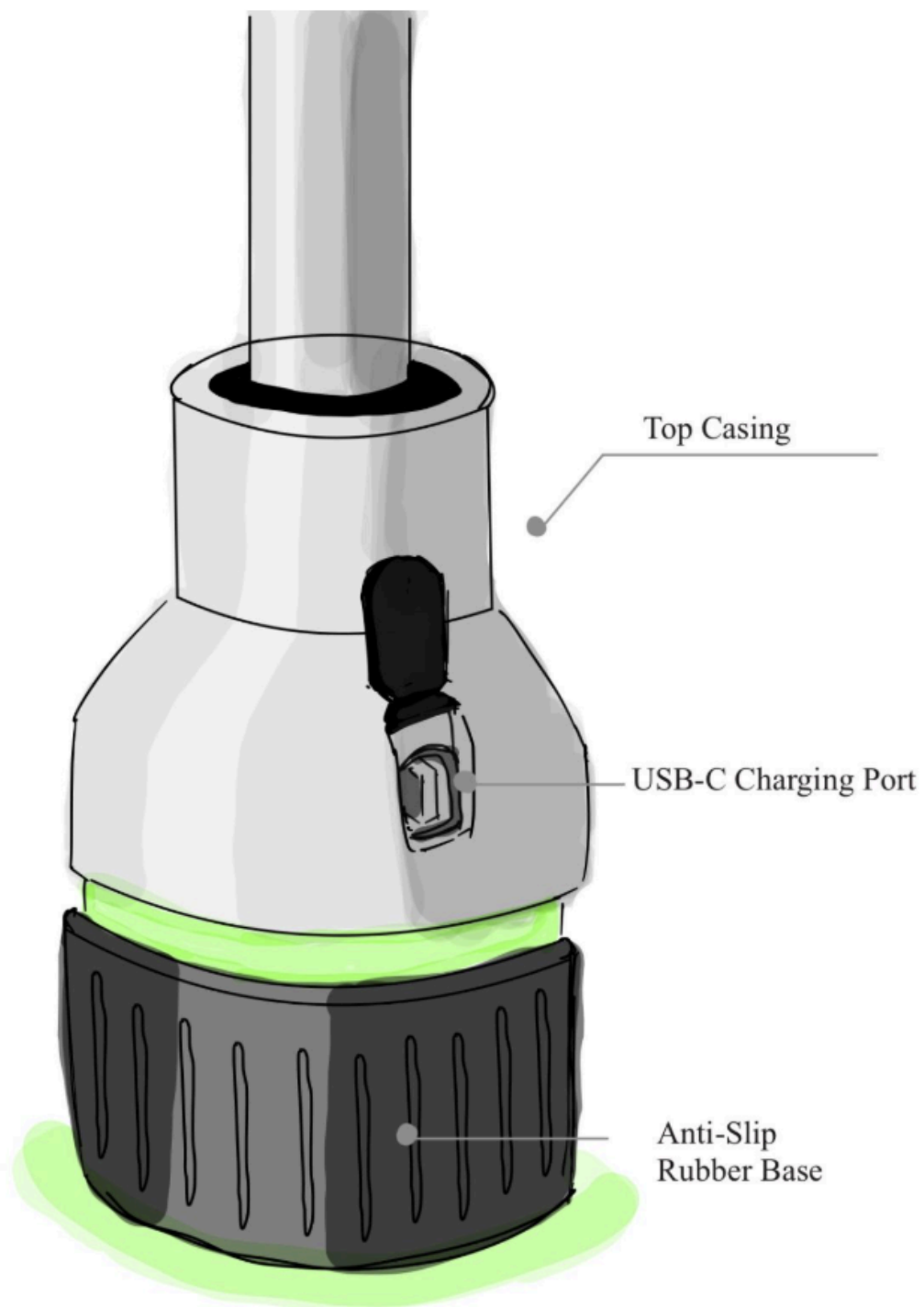
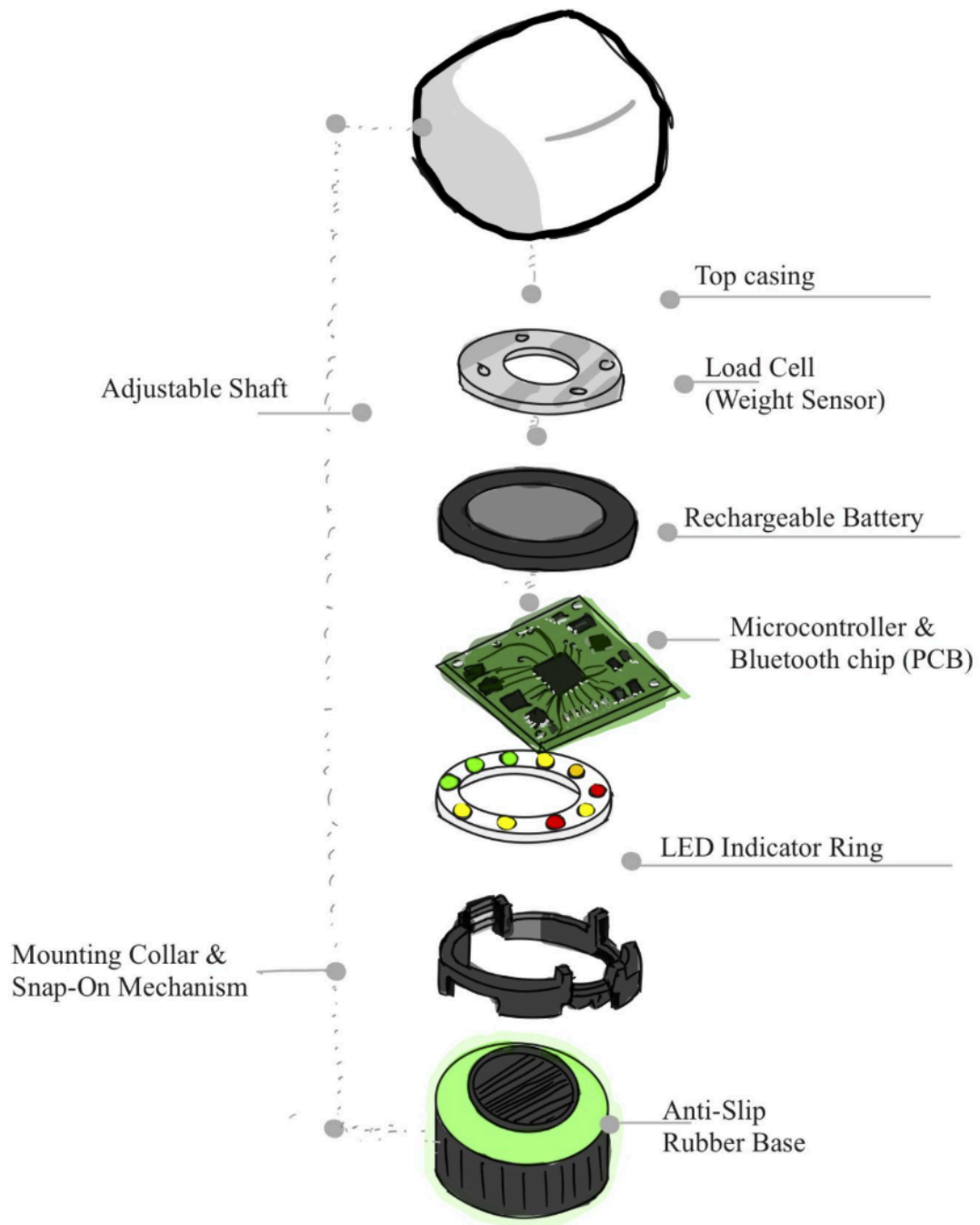


FIG. 3



FIELD OF INVENTION

[0001] The present disclosure relates generally to Assistive Technologies, including crutches, canes, walkers, and other orthopedic devices, more particularly to biomechanical sensing systems for load monitoring, gait analysis, and related signal processing.

BACKGROUND

[0002] The statements in this section merely provide background information related to the present disclosure and may not constitute prior art.

[0003] Assistive mobility devices such as crutches, canes, and walkers are commonly used during rehabilitation to help patients control how much weight they place on an injured limb. In many cases, physical therapists and physicians rely on limited in clinic observation and patient self reporting to evaluate recovery progress. Because most rehabilitation activity occurs outside of clinical environments, providers may have difficulty determining whether patients are consistently following prescribed weight-bearing guidelines. Existing rehabilitation practices have highlighted a need for more objective and continuous data collection between appointments. Without consistent quantitative measurements, tracking patient compliance and progression may be challenging. Additionally, current evaluation methods often lack direct measurement of ground reaction forces applied through assistive mobility devices during routine patient use.

[0004] Prior approaches include wearable activity trackers, rehabilitation monitoring applications, and gait analysis systems designed to measure movement or estimate physical activity levels. Some systems use instrumented insoles or external motion sensors to approximate weight distribution or track exercise compliance. While these approaches provide useful movement-related information, many rely on indirect measurements or require additional equipment separate from assistive mobility devices. As a result, accurate measurement of load applied through assistive mobility devices during routine use remains limited.

SUMMARY

[0005] As discussed above, existing rehabilitation methods often lack continuous and objective measurement of weight bearing during use of assistive mobility devices outside of clinical environments. It has been surprisingly discovered that integrating load measurement directly into a structural component of an assistive mobility device allows real time monitoring of patient weight bearing and improved communication between patients and healthcare providers.

One way to implement this is by using a crutch tip assembly that includes an embedded load cell configured to measure force applied between the crutch and a ground surface, and a wireless communication module configured to transmit load data to an external device. The system may provide real time feedback to a user when measured load exceeds or falls outside prescribed safe ranges. In some embodiments, this tool is when the data from the crutch tip is transmitted and may be displayed through a software

application to allow patients to track rehabilitation progress and to allow physical therapists or physicians to monitor recovery and adjust treatment recommendations.

The advantages of this system is that it can provide objective weight bearing data not typically available through standard rehabilitation methods, allowing for improved compliance monitoring and more informed recovery planning.

[0006] Further areas of applicability will become apparent from the description provided herein. It should be understood that the description and specific examples are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

DRAWINGS

[0007] The drawings described herein are for illustration purposes only and are not intended to limit the scope of the present disclosure in any way.

[0008] FIG. 1 is a system overview diagram of the smart crutch tip within an operational environment, illustrating the wireless connection between the smart tip, the user's mobile device, and the clinician dashboard via a bluetooth signal.

[0009] FIG. 2 is an exploded perspective view of the smart crutch tip, showcasing the internal components of the device, including a load cell (weight sensor), rechargeable battery, microcontroller and Bluetooth PCB chip, an LED indicator ring, and the mounting assembly. Green light indicates safety, yellow cautions the user, and red signals force overload.

[0010] FIG. 3 is a close-up perspective view of the smart crutch in an assembled state, highlighting the external casing, the USB-C charging port and its protective cover, the LED indicator ring, and the anti-slip rubber base.

DETAILED DESCRIPTION

[0011] The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or uses. It should be understood that throughout the drawings, corresponding reference numerals indicate like or corresponding parts and features.

[0012] This invention, referred to throughout as the "smart tip," is a pressure-sensing tip assembly designed to replace the standard rubber tip on assistive mobility devices like crutches, canes, and walkers. The smart tip works by embedding a load cell directly inside the tip housing, so that every time a patient takes a step and puts weight through their device, that force is measured, processed, and sent wirelessly to a digital interface, such a mobile or desktop application. This approach fills a real gap in rehabilitation care. Currently, clinicians mostly rely on patients self-reporting whether they are following weight-bearing instructions, and there is no objective way to verify that between appointments. The smart tip changes that by turning a device the patient already uses into a continuous data collection tool.

The load cell is the core sensing element of the smart tip and is the best method for directly measuring the force a patient applies through their mobility aid. It is positioned inside the housing along the vertical axis of the device, so it sits directly in the

load path between the crutch shaft and the ground. This placement allows it to capture the actual ground reaction force at the point of contact rather than estimating it from body motion or a separate wearable. A strain-gauge type load cell rated for 0–150 kg is preferred because it covers the full range of typical adult body weights used during partial and full weight-bearing rehabilitation. There are no prior approaches that directly measure force through the assistive device itself, which is the key advantage of this design.

The LED indicator ring is the best solution for giving the patient real-time feedback without requiring them to look at their phone mid-stride. The ring wraps around the exterior of the housing so it is visible during use, and it is controlled directly by the microcontroller based on a live comparison of measured force to the prescribed threshold. Green means the patient is within their safe weight-bearing range. Yellow is a caution that they are getting close to the limit. Red means they have gone over and need to reduce how much weight they are putting through the device. This three-color system is intuitive, requires no reading or interaction, and gives patients immediate information they can act on in the moment.

The microcontroller and Bluetooth PCB chip are the best solution for processing and transmitting the load data off-device. The microcontroller takes the raw analog signal from the load cell, converts it to a digital value, runs the threshold comparison logic, controls the LED ring, and sends the data over Bluetooth to the companion app. On the app side, patients can view their weight-bearing history and track progress over time.

Clinicians access the same data through a remote dashboard, which lets them monitor patient compliance, spot trends in recovery, and update treatment plans without needing the patient to come into the office. This two-sided interface is what makes the device valuable not just as a feedback tool but as a clinical monitoring system.

FIG. 1 shows the full operational picture of how the smart tip fits into a rehabilitation workflow. The smart tip is attached to the bottom of a crutch, and a Bluetooth signal connects it to the patient's smartphone running the companion app. That app then connects to the clinician dashboard, creating a direct data pipeline from the patient's daily activity to the healthcare provider. Every step the patient takes generates a force measurement that is logged in real time, giving clinicians a continuous record of how the patient is actually loading their injured limb throughout the day, not just during a 30 minute appointment once a week.

FIG. 2 breaks down the internal layout of the smart tip from top to bottom. Each layer of the assembly serves a specific function, and the components work together as follows:

- **Mounting Assembly:** The topmost component, located at the proximal end of the housing. It connects the smart tip to the shaft of the mobility device and includes an adjustable interface to fit different shaft diameters, making the device compatible with most commercially available crutches, canes, and walkers without modification.
- **Microcontroller and Bluetooth PCB Chip:** Positioned just below the mounting assembly inside the housing. This is the brain of the device, it receives the load cell

signal, runs the threshold logic, drives the LED ring, and handles all Bluetooth data transmission to the mobile app.

- **Rechargeable Battery:** A lithium-polymer (LiPo) battery that powers all active components. It connects to the microcontroller and charges through the USB-C port on the outside of the housing. A protective cover over the port keeps out moisture and debris during normal use.
- **Load Cell (Weight Sensor):** Centered within the housing along the vertical load axis. This is the force transducer that converts the compressive force of each step into an electrical signal. It connects directly to the microcontroller via wired leads.
- **LED Indicator Ring:** An annular ring of LEDs mounted around the outer circumference of the housing. It is wired to the microcontroller and displays green, yellow, or red depending on whether the measured force is within, near, or over the prescribed threshold.
- **Anti-Slip Rubber Base:** The bottommost component and the only part that contacts the ground. It is made from polyurethane rubber, which was selected for its grip on a variety of surfaces (tile, hardwood, concrete, and carpet) and its ability to absorb small shocks during walking.

FIG. 3 shows the smart tip fully assembled as it would appear in use. The outer casing houses all internal components and keeps them protected from everyday wear and environmental exposure. The USB-C charging port and its protective cover are visible on the side of the casing. The LED indicator ring is visible around the housing's outer edge. The polyurethane rubber base is at the bottom, making contact with the ground. The overall size and

shape of the assembled device closely mirrors a standard mobility aid tip so that it does not affect the patient's balance or change the feel of the device.

To assemble the smart tip, begin by placing the load cell in the center of the housing, oriented so its sensing axis aligns with the vertical load direction. Connect the load cell leads to the microcontroller PCB. Secure the battery adjacent to the PCB and wire it to the power input. Connect the LED ring to the microcontroller output. Attach the anti-slip polyurethane base to the distal end of the housing. Finally, secure the mounting assembly to the proximal end and seal the housing to protect all internal electronics from moisture and debris.

To use the device, the patient or a clinician slides the smart tip onto the bottom of the crutch or mobility aid using the mounting assembly. After powering the device on, the patient pairs it via Bluetooth to the companion mobile app. The clinician then sets the prescribed weight-bearing threshold in the app based on the patient's rehabilitation protocol. From that point on, the device runs automatically. The load cell measures force with every step, the microcontroller compares it to the threshold, and the LED ring gives the patient instant feedback. Load data are continuously transmitted to the app, logged over time, and accessible to the clinician through the remote dashboard. If the patient consistently exceeds their threshold, the app generates an alert so the clinician can follow up and adjust the treatment plan as needed.

The ground-contacting base is made from polyurethane rubber due to its combination of toughness, flexibility, and surface grip. The outer housing is fabricated from a rigid, lightweight polymer such as ABS or polycarbonate, both of which are impact-resistant and commonly used in medical device enclosures. The load cell is a strain-gauge type transducer with a rated capacity

of 0–150 kg to accommodate the full range of adult body weights during rehabilitation use. The power source is a lithium-polymer (LiPo) rechargeable battery, which offers a high energy density in a compact form factor suitable for integration within the tip housing. All dimensions are targeted to approximate those of a standard assistive device rubber tip to ensure the smart tip is a true drop-in replacement.

While this description focuses on a crutch tip, the same assembly can be adapted for use with canes, walkers, forearm crutches, and quad canes. The Bluetooth module can be swapped for other short-range wireless protocols like Wi-Fi if a different data pipeline is preferred. The LED indicator ring can be supplemented with an audible buzzer or a haptic vibration motor for users who may have difficulty seeing the light indicator. The load cell can be replaced with alternative force-sensing technologies such as piezoelectric sensors, force-sensing resistors (FSRs), or capacitive pressure sensors as long as they meet the required accuracy and load range. The USB-C charging port can be replaced with wireless inductive charging in future versions of the design.

Prior work in wearable activity monitoring and instrumented insoles has shown that continuous force and movement data can meaningfully improve rehabilitation outcomes. However, those systems estimate loading indirectly from body-worn sensors rather than measuring it at the point of contact with the assistive device. The smart tip builds on that foundation by integrating force measurement directly into the tip itself, producing device-specific ground reaction force data that prior approaches could not capture.

CLAIMS

What is claimed is:

1. An assistive mobility device tip assembly comprising:
 - a. A housing configured to attach to a distal end of an assistive mobility device
 - b. A load cell positioned within the housing and configured to measure force applied between the assistive mobility device and a ground surface
 - c. A microcontroller electrically connected to the load cell
 - d. A wireless communication module configured to transmit load data to an external device
 - e. A power source electrically connected to the microcontroller
 - f. A base member comprising an anti-slip material configured to contact the ground surface.
2. The assistive mobility device tip assembly of claim 1, further comprising a visual indicator that provides feedback based on measured load values.
3. The assistive mobility device tip assembly of claim 2, wherein the visual indicator comprises an LED indicator ring.
4. The assistive mobility device tip assembly of claim 1, wherein the power source comprises a rechargeable battery.
5. The assistive mobility device tip assembly of claim 1, wherein the wireless communication module comprises a Bluetooth transceiver.

6. The assistive mobility device tip assembly of claim 1, wherein the wireless communication module is configured to transmit load data to a mobile application.
7. The assistive mobility device tip assembly of claim 6, wherein the mobile application is configured to display load data for monitoring by a healthcare provider.
8. The assistive mobility device tip assembly of claim 1, wherein the housing includes an adjustable attachment interface configured to fit multiple assistive mobility device shaft diameters.
9. The assistive mobility device tip assembly of claim 1, wherein the microcontroller is configured to compare measured load values to a stored threshold.
10. The rehabilitation monitoring system of claim 1, wherein the software application is configured to generate an alert when the measured force exceeds a predetermined threshold.
11. The rehabilitation monitoring system of claim 7, wherein the remote computing system enables communication between the healthcare provider and the user based on the load data.

Ideation for Med Dev

- NOTE: Want idea to be novel/have novel concept, but does not necessarily NEED to be patentable
- Smart crutch tips
 - Provides feed back of how much weight is on the leg vs the crutch
 - Send info to app for instant feedback
 - healthcare providers can advise the patient on the best rehabilitation plan so that they heal quickly and correctly
- An AI-driven motion analysis system that supports physical therapists and athletes by objectively evaluating rehabilitation exercise form, repetitions, and range of motion to guide safe return-to-sport progression

2. ComeBack Mobility

Ukraine-based ComeBack Mobility wants to make it easy for patient care teams to monitor progress of the entire rehabilitation process after a leg injury.

Their Smart Crutch Tips measure how much weight a patient puts on their leg as they use crutches. Too much weight on their leg, and the crutch tips provide instant feedback so the patient can adjust accordingly. The crutch tips also pair with an app where patients can track their weight-bearing progress.

With this data, healthcare providers can advise the patient on the best rehabilitation plan so that they heal quickly and correctly. They can also send new rehab advice straight to the patient through the mobile app.

CEO Ilya Popov, the co-founder of ComeBack Mobility, was inspired to create this tool after going through his own injury. He raised the seed round of funding himself during the COVID-19 lockdown and gave an interview about maintaining a medical device startup during wartime.

(drawing actual sketches rn, but below is the basic design i ended up with. i tried to pick something as simple, cost effective, and innovative as possible) vv

Smart Crutch – Full Design

Lightweight • Adjustable • Sensor-Integrated Tip



SmartStride Crutch

AI-Driven Weight Monitoring & Rehab Guidance

Ergonomic Handle

- Pressure Sensors
- Grip Feedback

LED Indicator

Safe Caution Too Much

Smart Tip Close-Up

- FSR Sensors (x4)
- Microcontroller + BLE
- Rechargeable Battery
- LED Ring
- Rubber Base

2.4" diameter

Real-Time App Feedback



Live Feedback

- Green = Safe
- Yellow = Caution
- Red = Reduce Load



Vibration Alert

Clinician Dashboard



Adjustable Height 30" - 42"

- Lightweight Aluminum Shaft
- Durable
- Adjustable
- Foldable Clip (optional)

Smart Tip (Snap-On)

Force Sensors (Measure Weight)

Anti-Slip Rubber Base

- Shock Absorption
- 360° Grip

How It Works:

1. Step on Crutch → 2. Sensors Measure Load → 3. Instant Feedback → 4. App + AI Tracking →



Charging Port (USB-C)

Water-Resistant (IP65)